Chapter 1 Sets, Relations and Functions

"a set is many that allows itself to be thought of as a one"

Cantor

1.1 Introduction

The concepts of sets, relations and functions occupy a fundamental place in the mainstream of mathematical thinking. As rightly stated by the Russian mathematician Luzin the concept of functions did not arise suddenly. It underwent profound changes in time. Galileo (1564-1642) explicitly used the dependency of one quantity on another in the study of planetary motions. Descartes (1596-1650) clearly stated that an equation in two variables, geometrically represented by a curve, indicates dependence between variable quantities. Leibnitz (1646-1716) used the word "function", in a 1673 manuscript, to mean any quantity varying from point to point of a curve. Dirichlet (1805-1859), a student of Gauss, was credited with the modern "formal" definition of function with notation $y = f(x)$. In the 20th century, this concept was extended to include all arbitrary correspondence satisfying the uniqueness condition between sets and numerical or non-numerical values.

With the development of set theory, initiated by Cantor (1845-1918), the notion of function continued to evolve. From the notion of correspondence, mathematicians moved to the notion of relation. However even now in the theory of computation, a function is not viewed as a relation but as a computational rule. The modern definition of a function is given in terms of relation so as to suit to develop artificial intelligence.



Cantor
1845 - 1918

In the previous classes, we have studied and are well versed with the real numbers and arithmetic operations on them. We also learnt about sets of real numbers, Venn diagrams, Cartesian product of sets, basic definitions of relations and functions. For better understanding, we recall more about sets and Cartesian products of sets. In this chapter, we see a new facelift to the mathematical notions of "Relations" and "Functions".

Learning Objectives

On completion of this chapter, the students are expected to

- list and work with many properties of sets and Cartesian product;
- know the concepts of constants, variables, intervals and neighbourhoods;
- understand about various types of relations; create relations of any required type;
- represent functions in different ways;
- work with elementary functions, types of functions, operations on functions including inverse of a bijective function;
- identify the graphs of some special functions;
- visualize and sketch the graphs of some relatively complicated functions.